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REGULAR PAPER

A Knowledge Graph Based Disassembly Sequence Planning For End-of-Life Power Battery

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Abstract

The accurate and efficient intelligent planning of disassembly sequences plays a crucial role in ensuring the high-quality recycling of end-of-life power batteries. However, the solution space obtained by the metaheuristic algorithm is often incomplete, resulting in suboptimal sequence accuracy. Additionally, the complex and dynamic disassembly information associated with end-of-life power batteries poses challenges in analysis and reuse, leading to low efficiency in disassembly sequence planning. To address these issues, we propose a novel approach for planning disassembly sequences based on the knowledge graph representation of power batteries. Firstly, we construct an updateable and scalable disassembly information model using knowledge graphs to capture the dynamic information and assembly relationships among battery parts. Subsequently, we utilize a combination of topological sorting and backtracking algorithms on the constructed disassembly information graph to derive the optimal disassembly sequence. Finally, we demonstrate the feasibility and effectiveness of our approach through an illustrative case study involving an end-of-life power battery pack.

Keywords Disassembly sequence planning · Knowledge graph · End-of-life power battery · Knowledge reuse

1 Introduction

In response to global warming concerns, China has introduced the ambitious 'carbon peak' and 'carbon neutral' strategies. Notably, one significant hurdle hindering the nation's attainment of the 'double carbon' goal lies in the carbon emissions stemming from automobiles [6]. In 2020, China's new energy vehicle (NEV)sales accounted for a mere 5.4% of the total vehicle sales, exhibiting slower growth rates compared to European countries such as Germany and France. To address this issue, *The New Energy Vehicle Industry Development Plan (2021–2035)* has set an ambitious target for China to achieve NEV sales representing 20% of the total vehicle sales [11]. China has maintained its status as the world's largest producer and consumer of new energy vehicles for six consecutive years, with cumulative sales surpassing 5.5 million units [20]. The rapid growth of the electric vehicle market has driven the large-scale production of lithium-ion batteries. The exponential growth of discarded power batteries has led to a proliferation of battery categories and types. Secondary use of batteries and material recycling are two of the first options for disposing of waste batteries. Regardless of the chosen option, disassembly remains a pivotal step in both processes [3].

Disassembly sequence planning plays a key role in determining the optimal order and level of component disassembly in a product, aiming to maximize efficiency during the disassembly process. Accurate and efficient planning of disassembly sequences is crucial to enhance overall corporate revenues and facilitate intelligent recycling of large quantities of end-of-life power batteries. However, the prevalent use of heuristic algorithms, with their reliance on random solution approaches, often yield incomplete solution spaces and fail to ensure the accuracy of the disassembly sequence.

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Consequently, this can result in a loss of profit for the company. Furthermore, the intricate and dynamic nature of disassembly information related to end-of-life power batteries poses challenges in terms of reusability and analysis, thereby reducing the efficiency of disassembly sequence planning. Disassembly sequence planning for power batteries presents a fundamental challenge in representing the information and assembly relationships between battery components. Currently, the disassembly sequence planning for power batteries relies on a disassembly mixture graph to characterize the interconnections among battery parts. The disassembly matrix derived from the disassembled mixture graph is then used as input to the algorithm to determine the optimal disassembly time and sequence. However, as the number of end-of-life power batteries continues to grow, they exhibit a diverse range of categories and types. Unfortunately, the disassembly matrix focuses primarily on reducing assembly relations and lacks the capability to incorporate additional assembly relations. This limitation not only consumes significant time but also hampers the efficiency of power battery disassembly sequence planning.

By examining the disassembly information related to diverse categories and types of power batteries, a knowledge graph is employed to depict the intricate details of battery components and their assembly relationships. The dynamic attributes of end-of-life power battery disassembly information are meticulously processed through the knowledge graph's extensibility and updateability. Subsequently, the graph model is effectively solved using advanced graph theory algorithms, leading to the derivation of an optimal disassembly sequence. This innovative approach significantly enhances the efficiency of end-of-life power battery sequence planning, enabling streamlined and effective decision-making processes.

2 Literature Review

The problem of disassembly sequence planning for end-of-life power batteries has received significant attention from researchers, resulting in numerous studies addressing this issue. Notable contributions have been made by several researchers in this area. Zhan et al. [21] proposed a dual-objective disassembly sequence planning optimization model, to minimize the hazard index and energy cost during power battery disassembly operations. The predatory behavior of the goshawk is optimized by combining the characteristics of the disassembly sequence planning problem to improve its performance. Xiao et al. [17] proposed a dynamic disassembly Bayesian network method based on the disassembly graph model of electric vehicle batteries. Manufacturers provides dynamic process optimization to infer the optimal disassembly sequence using the forward–backward algorithm and the Viterbi decoding algorithm, which are used to solve uncertain disassembly structure representation and optimal disassembly sequence selection for end-of-life power batteries. Cong et al. [5] considered multi-objective optimization factors including the profit, workload ratio, quantity of disassembly workstations, energy utilization, and safety in battery disassembly workstations and developed a hybrid algorithm that combines genetic algorithm and fireworks algorithm. Guo et al. [9] proposed a lexicographic multi objective scatter search method to solve the proposed multi objective optimization problem, and established a novel multi objective optimization model such that the energy consumption and disassembly time are minimized while disassembly profit is maximized. Yin et al. [19] simulated the process of obtaining the best individual in a population and utilized a two-population genetic algorithm to identify optimal solutions within a mathematical model for used mobile phone disassembly. Mao et al. [12] presented a multi objective optimization model for the disassembly sequence of end-of-life car parts under uncertain conditions, combining a stochastic optimization algorithm combined with artificial intelligence technology. Cong et al. [4] proposed a high-yield EOL strategy method, making decisions regarding optimal disassembly sequence, levels and EOL options for components. This approach extends the non-destructive disassembly matrix to a disassembly transition matrix that also takes into account potential options for components, as well as transition costs between disassembly operations, tool costs, and recovery regulations or objectives. Guo et al. [8] proposed a model to optimize selective disassembly sequences subject to multi-resource constraints for maximize disassembly profit by using two scatter search algorithms with different combinatorial operators.

By analyzing the disassembly sequence planning process described above, this work finds that the extraction of product disassembly features and the expression of disassembly information affect the efficiency of disassembly sequence planning. In order to transform informative data such as battery structure into quantifiable data. Xu et al. [18] developed a disassembly model employing the disassembly stability matrix, fastener priority matrix, fastener component matrix, and component motion priority matrix to represent constraint and connection relationships between product parts. Rickli and Camelio [14] derived disassembly feasibility cost, and environmental impact from cost matrices and environmental impact matrices. Tao et al. [15] introduced an automatic self-decomposed disassembly precedence matrix. While matrix-based approaches provide disassembly sequences, researchers with limited empirical knowledge may struggle to understand the implications of these matrix-based constraint relationships. Hence, it is challenging to directly represent the battery structure through the disassembly matrix. In an effort to convey product disassembly information and knowledge in a more intuitive and concise manner, researchers have increasingly turned to graph-based models to characterize the constrained relationships among power battery parts. Ren et al. [13] proposed an asynchronous parallel disassembly planning method utilizing a disassembly hybrid graph to represent connection relationships and priority constraints among product parts. In a similar vein, Tian et al. [16] employed AND/OR graph diagrams, in conjunction with constraint matrices, priority matrices, and continuity matrices, to generate feasible disassembly sequences. To enhance the clarity and comprehensiveness of prioritized disassembly and connection relations.

However, the above studies only elaborate on the method of expression of the disassembly information and do not analyze the disassembly information itself, which can lead to the underutilization of a significant amount of the disassembly information and hence the reduction of the efficiency of disassembly sequence planning. Efficient utilization of product disassembly information and fast extraction of disassembly features are beneficial for improving the efficiency of disassembly sequence planning. Therefore, to enhance the efficacy of power battery disassembly sequence planning in this study, it is imperative to develop a dynamic and adaptable disassembly information model. As such, this study delves into the disassembly sequence of power batteries and presents an efficient planning method for power battery disassembly sequence based on knowledge graph. A scalable and portable power battery disassembly information model is developed by using knowledge graphs to describe the relationships of the internal parts of the power battery, and an algorithm is used to solve the graph model.

The main contributions of this paper are as follows:

(1) A combination of topological ordering and backtracking algorithms is used to solve the graph-theoretic problem of selecting the optimal disassembly sequence among all possible disassembly sequences.
(2) A knowledge graph-based disassembly information model of a power battery is constructed to describe the internal parts of the power battery and their connection relations.
(3) The model is proposed for updateability and scalability to handle dynamic power battery disassembly information.

3 Disassembling Sequence Planning Based on Knowledge Graph

3.1 Construction of Disassemble Information Model Framework

Disassembling sequence planning based on knowledge graph of the framework diagram is shown in Fig. 1. The proposed approach consists of four key components. Firstly, the disassembly characteristics of end-of-life power batteries, including battery type, structural information, disassembly time, and tools utilized, are extracted. Unstructured knowledge is then assimilated into novel information features to augment the existing disassembled knowledge graph. Secondly, through the process of structuring and distributing these information features, the disassembly information knowledge graph of the end-of-life power batteries is generated and stored as historical data for future reference. Subsequently, with the objective of minimizing the disassembly time, a graph-based traversal algorithm is employed to solve the disassembly information knowledge graph and optimize all feasible disassembly sequences. Finally, the disassembly sequence yielding the shortest disassembly time is obtained as the optimized solution.

We examine the optimal disassembly sequence for end-of-life power batteries and present a disassembly information model that captures the knowledge and information associated with power battery disassembly, including disassembly tools, battery structure, degree of disassembly, disassembly time, and constraint relations among parts. The assumptions underlying the model are outlined first.

- The power battery disassembly planning is from the battery pack disassembly to the battery module, without disassembling the battery module.
- The disassembly process is continuous, ignoring worker and machine rest periods.

There are four constraints on the model. Constraint (1), (2) and (3) are provided to avoid the disassembly information model to be an empty model, constraint (4) is generated by the relationship between product parts.

$$Card(D) \geq 0 \tag{1}$$

$$Card(M) \geq 0 \tag{2}$$

$$F \geq 0 \tag{3}$$

$$\sum_{i=1}^{F} \lambda_{Ii} = \sum_{i=1}^{F} \lambda_{Oi} \tag{4}$$

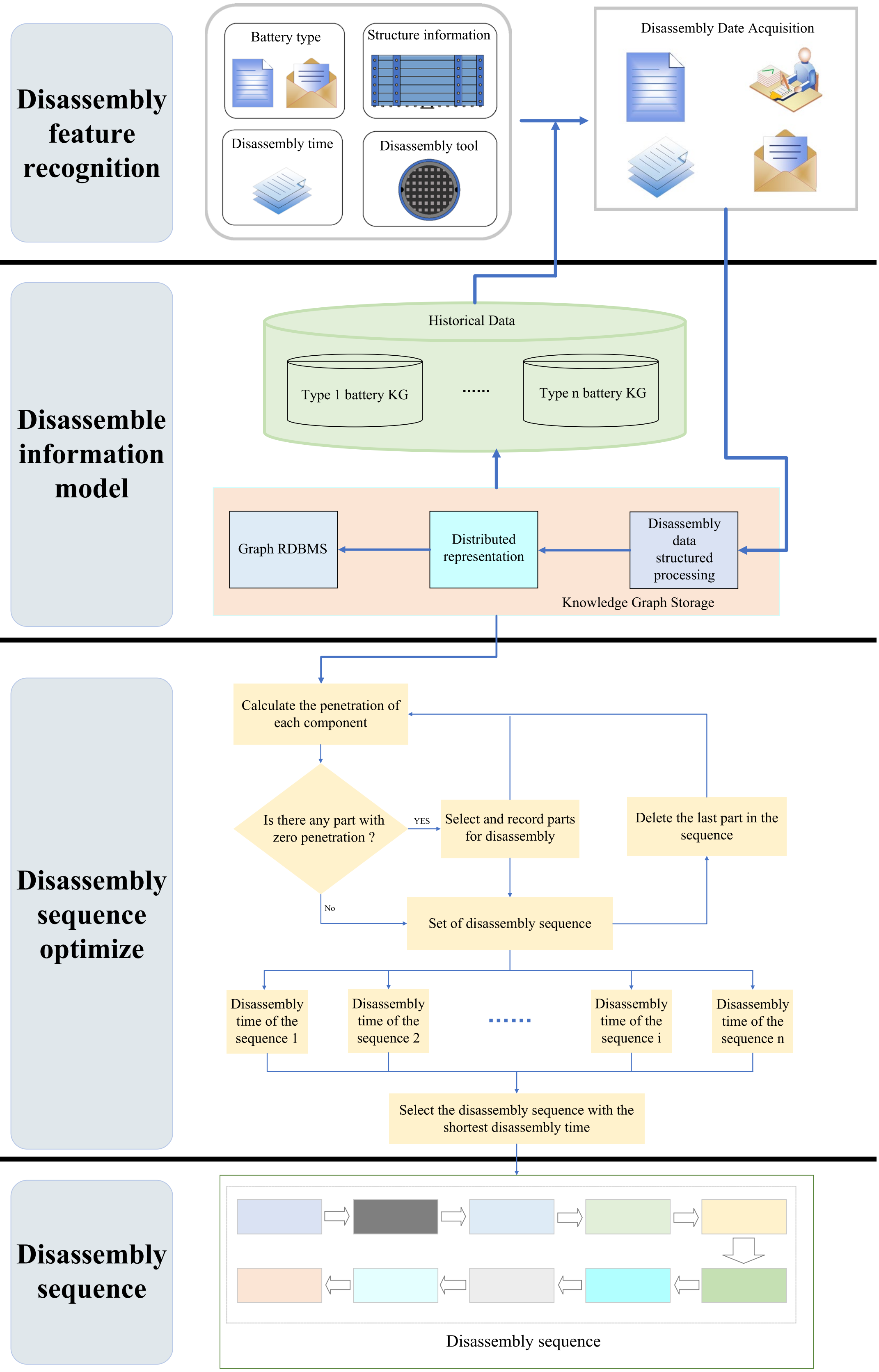


Fig. 1 Framework of power battery disassembly model based on knowledge graph

Card(X) the number of set X,D Set of disassembled parts, M Set of non-disassembled parts, F Total number of parts, λ_{Ii} The "in-degree" of part i, λ_{Oi} The "out-degree" of part i, i The number of part.

The concept of 'in-degree' refers to the count of parts directly constraining a given part i, while 'out-degree' denotes the count of parts directly constrained by part i. Formulas (1), (2) and (3) are employed to guarantee the absence of zero parts within the product, ensuring the meaningfulness of the entire disassembly sequence. Formula (4) ensures that the sum of 'in-degree' and 'out-degree' for all parts of the product remains equal, thereby validating the accuracy of the established information diagram model.

Input data for the model:

$$F\ I\ O\ A$$

Output data for the model:

$$S\ P$$

I Set of "in-degree" of parts, O Set of "out-degree" of parts, A Set of Parts disassembly time, S Optimal disassembly sequence,P Time of sequence S.

The model takes as input the structural parameters of the end-of-life power battery, including the sets of 'in-degree' and 'out-degree' for each part, the total number of parts, and the disassembly time associated with each part. The output of the model consists of the optimal disassembly sequence and the corresponding disassembly times.

To address the output requirements of the model, it can be divided into two main components: determining the optimal disassembly sequence and estimating the disassembly time for this sequence. In this model, the criterion to establish the optimal disassembly sequence is based on the disassembly time of the product. Thus, the formation of the optimal sequence is achieved by the following method.

The disassembly time of a power battery consists of two parts, one is the time required to disassemble the part, and the other is the tool change time and other times, which include tool movement time, fixture fixing time, etc. Such as formula (5)

$$T_i = T_{1i} + T_{2i}, \mathrm{When} Card(M) = 0 \tag{5}$$

$$W_i = P - T_i = \begin{cases} > 0,\ S\, remains \\ = 0,\ S\, remains \\ < 0,\ S = K_i \end{cases}$$

$$i = 1, 2, 3, \ldots \ldots, n \tag{6}$$

T_{i} Disassembly time of sequence i, T_{1i} Disassembly time of parts in sequence i, T_{2i} Disassembly time of tool change and others in sequence i. In comparison to P, T_i refers to the disassembly time of sequence i, where P represents the disassembly time of the optimal disassembly sequence. n The number of feasible disassembly sequences, K_{i} The disassembly sequence of i, W_i Transformation index, its purpose is to change the optimal disassembly sequence by the magnitude of its value. When $W_i \geq 0$, S remains the original sequence: When $W_i < 0$, then replace K_i with S, and so on. When $i = n$, S is the disassembly sequence with the shortest disassembly time among all disassembly sequences of the product. Equation (5) represents the calculation of the disassembly time for each generated feasible sequence. By applying this equation, the disassembly time is determined for each potential sequence. Furthermore, Eq. (6) embodies the optimization process wherein the disassembly sequence is refined based on the associated disassembly time, ultimately resulting in the identification of the optimal disassembly sequence.

3.2 Disassembly Information Representation of the Model

The effective utilization of knowledge graphs in planning disassembly sequences for power batteries relies on the ability to express and update assembly relations and disassembly information promptly using graph frameworks, thereby optimizing the disassembly sequence.

In the analysis of battery structure information, the formation of constraints among battery parts necessitates the organization of this knowledge. To achieve this, a knowledge graph is employed to organize and analyze the disassembly knowledge information while satisfying the triadic requirements of constraint relations between parts and components. The fundamental unit of a knowledge graph is a triad [2], [10], composed of an entity-relationship-entity composition, where the relationship links two entities. In this study, the connected parts of the power batteries are represented as entities within the knowledge graph, and the constraints and connections between these parts are represented as relations. This part-constraint-part structure constitutes the disassembly triad of the power battery, forming a knowledge graph representing the disassembly information of the end-of-life power battery. Furthermore, the disassembly time and tools associated with the power battery pack are considered as properties of the parts. Leveraging the power of knowledge graphs, we constructed a graph-based representation to capture the structural information of the battery.

The aforementioned discussion focused on illustrating the disassembly relationships and constraints between parts within the established power battery disassembly information model. However, knowledge graph based disassembly information models also incorporate other relevant information in various ways. This includes the properties associated with each part of the power battery, such as the required disassembly tool, the number of parts, and the time required to disassemble an individual part. These additional properties are effectively represented within the knowledge graph and thus become part of the input data for the algorithm, alongside the connections and constraints between parts.

In the comprehensive knowledge graph, the nodes and edges play a vital role. The nodes are categorized as green and red, where green nodes represent functional parts, and red nodes symbolize connecting parts [1], [10]. The directed edges between two nodes signify the constraint relationship between the parts. Specifically, the part indicated by the pointing node is constrained by the part represented by the indicated node, and the disassembly sequence of the parts must adhere to the direction of the arrows within the knowledge graph. From the preceding analysis, it can be inferred that considering the constraint relationship between parts alone, a part can only be disassembled when there are no arrows pointing towards the node representing that part. Similarly, during the formation of the disassembly sequence, the release of the binding relationship between a part and other parts corresponds to the disassembly of each part. This transformation is manifested in the knowledge graph by removing the outgoing arrows from the node representing the disassembled part. Consequently, during the generation of the disassembly sequence, if no arrow points to a particular node, that part is deemed removable. Once a part is disassembled, all arrows originating from the node representing that part are also eliminated.

The components within an end-of-life power battery can be classified into two distinct types based on their functionalities: functional parts and connecting parts. Functional components, such as battery modules and copper wire series, play a crucial role in the operation and performance of batteries. On the other hand, the connecting parts, including bolts and wire harnesses, serve to secure and support the functional parts. Since there is no direct connection between functional parts, they are linked through covering. When a functional part is removed, at least one functional part is exposed, which can then be detached following the same direction as the previously removed functional part. The initial disassembly direction of functional parts can be viewed as the disassembly direction of the initial functional parts. However, in the case of typical battery packs, the initial functional parts are usually external parts such as the top cover. These can be disassembled following the commonly used disassembly direction for removing the top cover. As a result, the initial disassembly direction of functional parts can be determined. Consequently, the disassembly direction of functional parts holds significant importance, and attention should be solely focused on the connection and covering relationships between the parts. In addition, the knowledge graph undergoes updates by incorporating new battery disassembly information into the original knowledge graph. By comparing the original and new disassembly information, the differences are identified, the redundant nodes, constraints, and attribute values in the original knowledge graph are removed, while the unchanged disassembly information is retained. the newly acquired disassembly information is added to generate an updated disassembly knowledge graph.

3.3 Model Solution and Formation of Disassembly Sequence

Due to the intrinsic properties of the end-of-life power battery structure, the two parts cannot be constrained in a mutual way. As a result, the knowledge graph constructed from the disassembly information of the end-of-life power battery is classified as an acyclic graph. The entire knowledge graph exhibits orientation due to the presence of constraint relations between parts. Consequently, the resulting disassembly information knowledge graph is a directed acyclic graph of a specific type. Topological sorting is commonly employed to find sequences in directed acyclic graphs; However, it does not give rise to all feasible sequences. To overcome this limitation, this study uses a combination of topological ordering and backtracking algorithms to solve directed acyclic graphs for disassembly knowledge graphs.

Topological sorting, a well-established technique, is employed to arrange all vertices in a directed acyclic graph into a linear sequence, guaranteeing that each node appears only once [7]. However, obtaining all feasible sequences solely through topological sorting poses challenges, necessitating the incorporation of a backtracking algorithm into the knowledge graph. This combination enables the retrieval of all feasible sequences within the knowledge graph, facilitating the computation of the shortest disassembly time for each sequence.

During the disassembly process, an iterative approach is employed to determine the optimal sequence. Initially, an unconstrained part is selected and recorded, followed by its removal. Subsequently, another unconstrained part is selected, repeating this iterative process until all parts have been disassembled. This yields a disassembly sequence along with the corresponding disassembly time. Subsequently, another unconstrained part is selected, and this iterative process is repeated until all parts have been disassembled. By calculating the disassembly time for all feasible disassembly sequences, the disassembly sequence with the shortest time is ultimately identified as the optimal solution.

During the disassembly process, it is crucial to determine the removable state of each part. Parts in a removable state are considered as candidates for subsequent disassembly steps. At each step in forming the disassembly sequence, a part is selected from the available candidate parts. If multiple candidate parts are present, a selection is made and a new set of candidate parts is formed by incorporating the remaining candidates. This iterative process continues until no candidate parts are available for selection, resulting in the completion of a disassembly sequence. In order to obtain all possible disassembly sequences, it is necessary to keep track of the selected parts among the candidates. When forming the next disassembly sequence, previously selected parts are excluded to allow for the selection of other removable parts. This approach enables the generation of multiple disassembly sequences. However, to ensure that the disassembly sequence is unique and complete, backtracking is employed, which allows exploration of all possible disassembly sequences without duplication or omission.

In this model, the formation of the product's disassembly sequence relies on the dynamics of the "in-degree" and "out-degree" of its internal parts. The decision to disassemble a particular part is based on its "in-degree" value. When the "in-degree" of a part reaches zero, it signifies that the part is ready for disassembly and can be added to the ongoing disassembly sequence. Consequently, the "out-degree" of the part is updated to zero. Following the constraint defined by Eq. (4), the reduction in "out-degree" is proportional to the decrease in the corresponding "in-degree". Subsequently, the "in-degree" of each part is reevaluated. If there are multiple parts with an "in-degree" of zero, one part is selected from the available options. This process continues iteratively, allowing the construction of a complete disassembly sequence. The disassembly sequence is obtained using the following formula:

Since the product is removable, then:

$$\exists j \in N, N > 0, \ then \lambda_{Ij} = 0 \tag{7}$$

The meaning of Eq. (7) is to find current battery parts for disassembly, and j represents the number of parts available for disassembly.

This study, the following definition is given:

$$K_i + j = K_i \cup \{j\} \tag{8}$$

$$K_i - j = K_i - \{j\} \tag{9}$$

Definitions (8) and (9) define $k_i + j$ and $k_i - j$ for subsequent calculations.

This definition leads to:

$$K_i = K_i + j \tag{10}$$

The meaning of Eq. 10 is to add the part j to the disassembly sequence K_i.

This is repeated to obtain the disassembly sequence of i. After the disassembly time of this sequence is calculated, the following backtracking operation is performed.

$$K_i = K_i - j \tag{11}$$

Using formula (11) to go back to the place where the last selection was made and select another part to get a new disassembly sequence again. By iteratively applying Eqs. (10) and (11) in this manner, it is possible to obtain all feasible disassembly sequences. Additionally, the disassembly time for each feasible sequence can be calculated using Eqs. (5) and (6), allowing for the determination of the optimal disassembly sequence and the corresponding time required for disassembly according to that sequence. This iterative process ensures comprehensive exploration of possible sequences and facilitates the identification of the most efficient disassembly solution based on the disassembly time criteria.

4 Case Study and Discussion

In this study, a power battery pack obtained from a recycling enterprise in Wuhan was used as the primary study subject. The battery pack consists of a ternary group material and encompasses various components, including 20 battery modules, a battery management system (BMS), a high-voltage box, essential restraint devices, and power electronics. Figure 2 presents a visual representation of such a power battery system. Due to the specialized disassembly tools required, such as cutters, and the potentially hazardous nature of the batteries, improper handling can lead to dangerous accidents. Hence, this work focuses solely on the disassembly of the battery pack into battery modules, while the disassembly of the modules into battery units has not been extensively examined. By categorizing the power battery and utilizing its structural information, a knowledge graph representing the disassembly information of the power battery is generated. The internal parts and assembly relations of the battery are visualized using the Neo4j software, as demonstrated in Fig. 3.

As illustrated in Fig. 3, the depicted knowledge graph highlights the structural properties of the power. The green nodes within the graph symbolize the functional parts of the power battery, whereas the red nodes represent the connecting parts. The connections shown between the green and red nodes exemplify the constraints associated with the relationship between the functional part and the connecting part. Similarly, the covering constraints depicted between the green nodes indicate relations and constraints between different functional parts.

As can be seen from Fig. 3, the power battery has a total of 39 parts, including 20 functional parts and 19 connecting parts. The power battery has a total of 401 parts, excluding gaskets, film, etc. in the power battery pack.

The assumptions used in the calculation of this study are: the form of labor is the disassembly form of labor workers using tools to disassemble the power battery; the

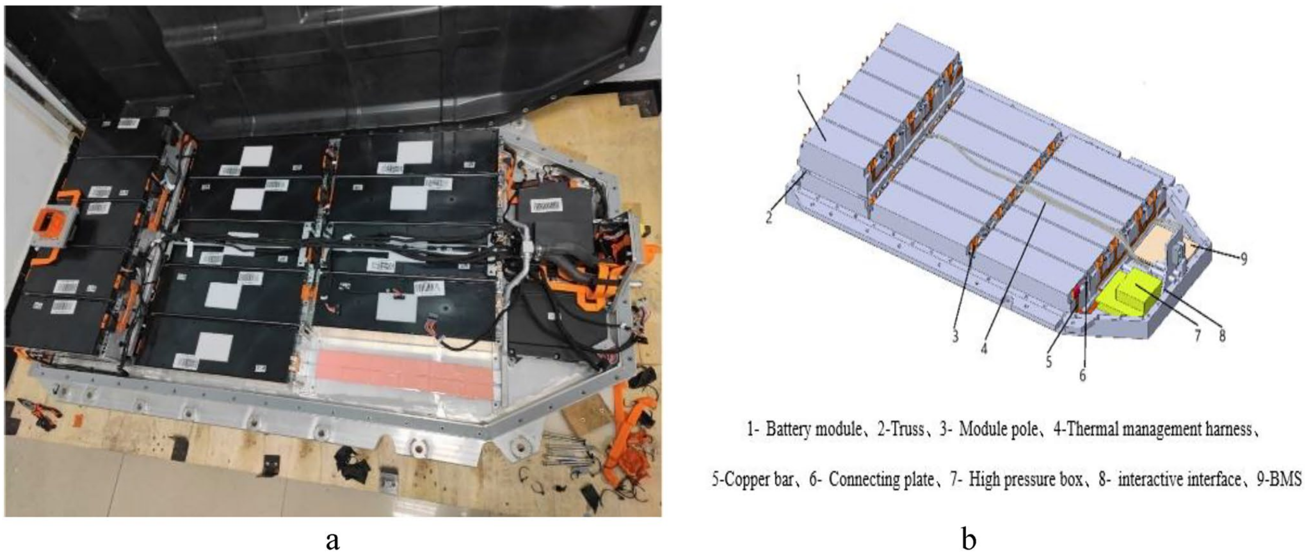


Fig. 2 General picture of power battery structure

number of labor workers is 1; the time of labor workers in the process of moving and switching disassembly tools is fixed at 6 s in total; The fixed time for other activities (including fixture time and tool movement time) is set at 2 s; the disassembly time varies for different components; disassembling the same component takes the same amount of time; complete disassembly of this power battery; disassembly of the battery pack into battery modules; and non-destructive disassembly.

Some of the input data related to this case algorithm is:$N = 39$. The relationships of the parts also into the algorithm. The topological sorting and backtracking algorithm are used to obtain a disassembly sequence with the lowest disassembly time among all feasible disassembly sequences, and the final results are obtained after calculation by the algorithm, of which some feasible solutions are shown in Table 1.

In the provided table, we can observe that the first column represents the parts of the power battery. The second column displays the disassembly sequence of Sequence 1, where the parts are arranged in ascending order based on their numerical values. Similarly, the third column represents the disassembly sequence of Sequence 2, and the last column represents the disassembly sequence of Sequence 3. The final row of the table presents the corresponding disassembly time required for each sequence. The sequential disassembly time for Sequence 3 is 1657.6 s, while Sequence 1 requires 1663.6 s. The reason for this outcome lies in the variations in tool change and tool movement times among different sequences, primarily stemming from differences in the number of tool changes across sequences. This discrepancy in disassembly times arises from the distinct disassembly sequences, resulting in variations in disassembly times.

Based on the obtained results, all feasible disassembly sequences and their corresponding times have been obtained. By applying algorithmic optimization, the optimal disassembly sequence and the corresponding disassembly time can be determined. Figure 4 illustrates the process of algorithmic optimization search. The disassembly time for this particular sequence is found to be 1657.6 s.

Figure 4 represents the process of finding the optimal disassembly sequence according to the disassembly time. The horizontal coordinate in the figure is the number of disassembly sequences and the vertical coordinate is the shortest disassembly time among these sequences. It can be shown in Fig. 4 after the number of disassembly sequences reaches 300, the shortest disassembly time has been kept constant, indicating that the optimal disassembly sequence has emerged among the first 300 sequences.

The above conclusion only verifies the possibility of power battery disassembly sequence based on knowledge graph, and the efficiency of its method is to be verified. This study verifies the validity of the study by calculating the time required to obtain the disassembly sequence in different ways and comparing it with the method proposed in this study.

Now we face an end-of-life power battery pack and the time efficiency of the time savings obtained its disassembly sequence through the knowledge graph over others as follows. Equation (12) is used to calculate the efficiency of the disassembly sequence obtained.

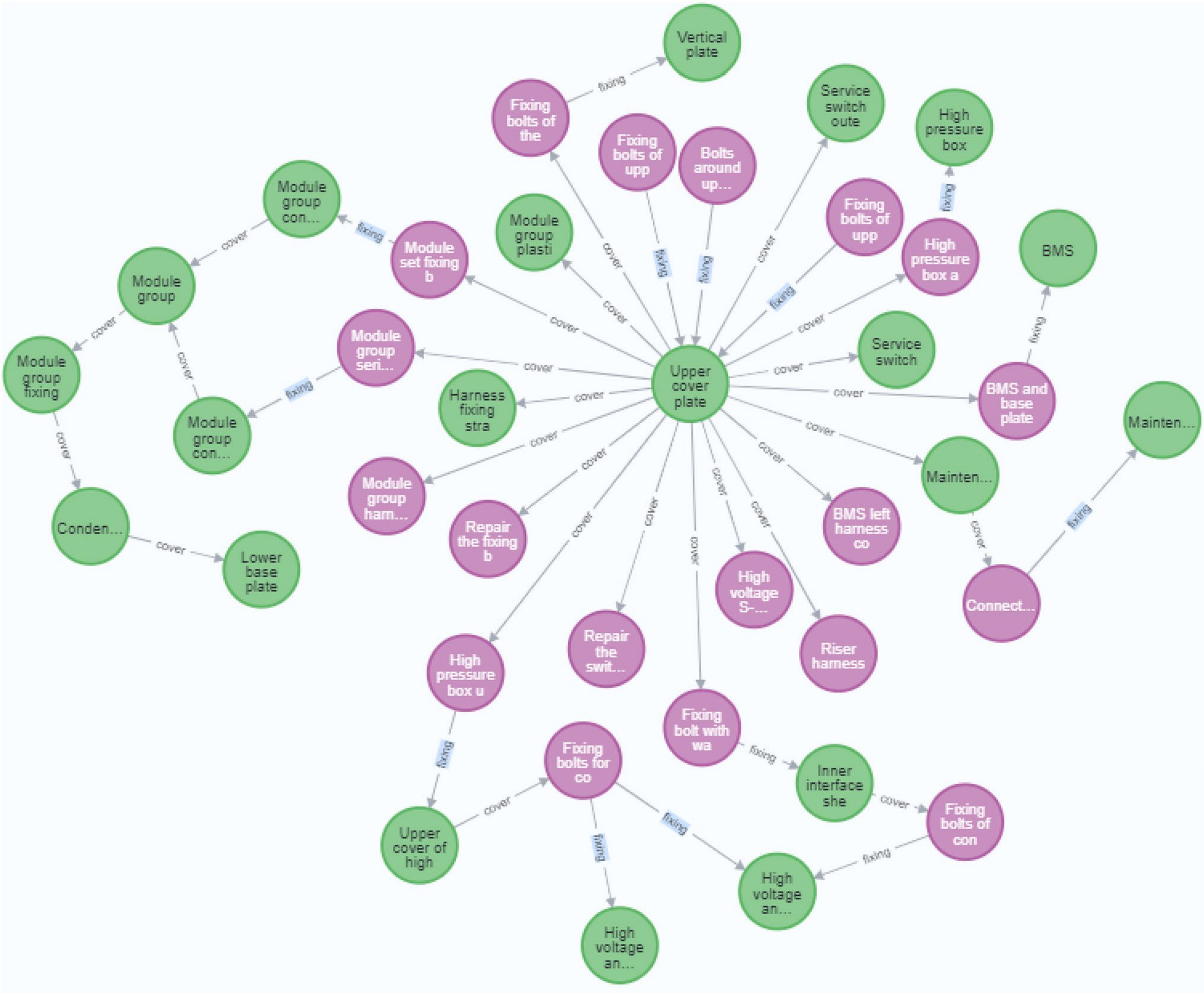


Fig. 3 Knowledge graph of power battery disassembly

$$\alpha_x = \frac{\sum_{j=1}^{3} t_{xj} - \min\left(\sum_{j=1}^{3} t_{ij}\right)}{\min\left(\sum_{j=1}^{3} t_{ij}\right)} (i = 1, 2, 3; x = 1, 2) \tag{12}$$

α_x is the time efficiency compared to other methods,t_{ij} is the time required for different disassembly steps in different disassembly methods, i is for different disassembly methods, j is for different disassembly steps.

In this case, the following results are part of the data obtained by running on a PC with AMD Ryzen 5 5600H 3.30 GHz CPU and 16 GB RAM under Windows 11 operating system:

The schedule of battery disassembly sequence planning using different methods is depicted in Fig. 5. The blue bars represent the time required to construct the disassembly mixture diagram using various planning methods. The green bars represent the time required to build the disassembly constraint matrix using different planning methods. The orange bars represent the time required to find the optimal disassembly sequence using algorithms in different planning methods. Finally, the red bars indicate the total time required using different planning methods.

This study aims to compare the time required for three different methods of disassembly sequence planning for end-of-life power batteries. The first method involves forming a disassembly matrix using a disassembly hybrid graph and then employing a meta-heuristic algorithm to obtain the disassembly sequence. The second method entails directly constructing the disassembly matrix of end-of-life power batteries by experienced workers and using an algorithm to obtain the disassembly sequence. The third method involves

Table 1 Partial feasible solutions of the algorithm

Product components	Sequence 1	Sequence 2	Sequence 3
Bolts around the upper cover	1	2	3
Fixing bolts of upper cover and vertical plate	2	1	1
Fixing bolts of upper cover and maintenance switch	3	3	2
Upper cover plate	4	4	4
Module group plastic buckle	5	15	9
Module group series copper bar fixing short bolt	6	17	12
High pressure box upper cover fixing bolt with washer	7	5	7
BMS and base plate fixing bolts	8	18	18
Fixing bolts of the outer interface of the vertical plate	9	11	20
Fixing bolt with washer at inner interface of vertical plate	10	6	6
Repair the fixing bolts of the outer ring cover of the switch	11	9	10
Module set fixing bolts	12	10	11
Service switch outer ring cover	13	24	16
Maintenance switch bottom cover	14	12	14
Harness fixing strap	15	16	5
Module group fixing plate	16	33	25
Module group connection board	17	20	28
Inner interface shell of vertical plate	18	7	8
Fixing bolts of connecting copper bar inside the vertical plate	19	23	22
Connecting bolts and nuts of maintenance switch and series copper bar	20	21	24
High voltage box S-box harness connector	21	35	17
Repair the switch interlock harness connector	22	27	34
Upper cover of high pressure box	23	19	15
BMS left harness connector	24	14	27
High pressure box and base plate fixing bolts	25	8	21
Fixing bolts for connecting copper bar in high-voltage box	26	22	23
High voltage box and module connection copper bar	27	32	29
Maintenance switch series copper bar	28	34	33
Module group harness connector	29	31	32
High voltage box and vertical plate connecting copper bar	30	13	13
Riser harness	31	25	19
Service switch	32	30	35
BMS	33	26	37
Vertical plate	34	28	30
Module group connecting copper bar between modules	35	29	26
High pressure box	36	36	31
Module group	37	37	36
Condensing tube	38	38	38
Lower base plate	39	39	39
Disassembly time	1663.6 s	1669.6 s	1657.6 s

building a disassembly information graph model and utilizing a graph theory-based algorithm to derive the disassembly sequence.

The results from the graph indicate that the time required for the third planning method is shorter than that of the first two planning methods. Detailed data calculations reveal that the efficiency of the sequence obtained by the third planning method is 14.78% higher than that of the first planning method and 10.71% higher than that of the second planning method. This confirms the effectiveness of the method proposed in this study.

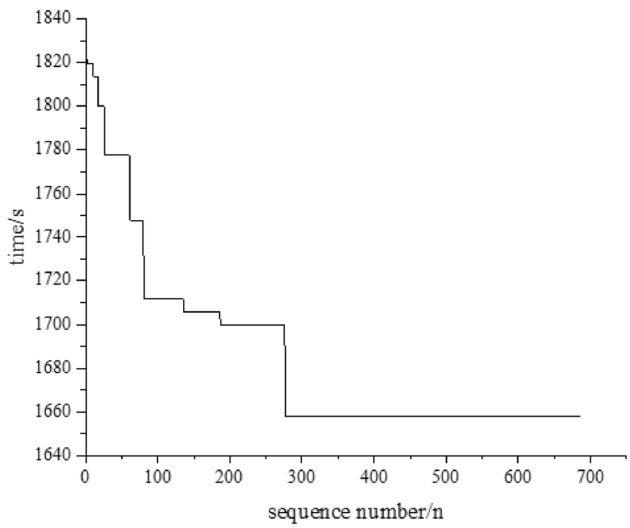


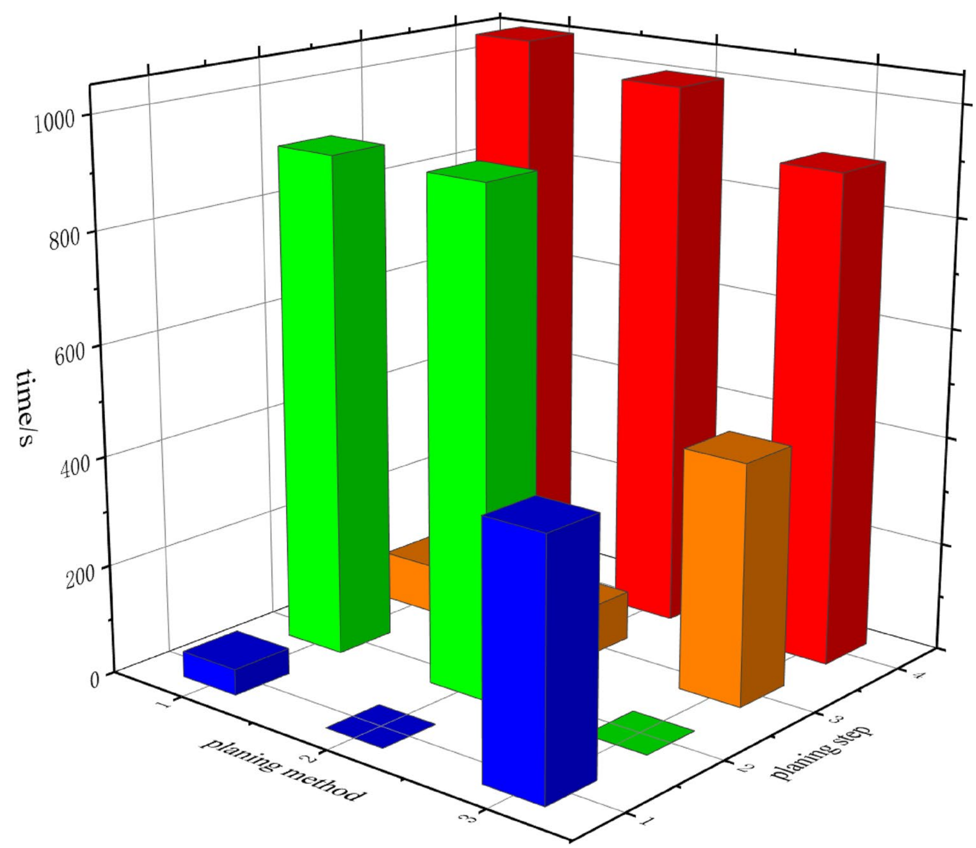
Fig. 4 Diagram of the algorithmic optimization search process

5 Conclusion and Future Work

To address the inefficiency in planning power battery disassembly sequences due to the diverse categories and types of power batteries and the abundance of disassembly information, this work proposes an efficient knowledge graph-based power battery disassembly sequence planning method. The proposed method is designed to enhance the efficiency of planning disassembly sequences for power batteries with complex internal structures. To address the challenges posed by multiple types and models of end-of-life power batteries, this approach utilizes knowledge graphs to represent all components within a power battery and their corresponding connectivity and constraint relations. By employing this representation, the disassembly information graph of the power battery is constructed, which serves as input data for the algorithm. The algorithm combines topological sorting and backtracking techniques to determine the optimal disassembly sequence. Using a specific power battery as an illustrative example, the proposed method intelligently generates a disassembly sequence with a minimum disassembly time of 1657.6 s. This result highlights the effectiveness of the method in achieving efficient power battery disassembly sequence planning.

Its efficiency is enhanced by 14.78 percent compared to the approach involving a combination of disassembled mixture graphs, disassembly matrices, and algorithms. Moreover, it demonstrates a 10.71 percent improvement in efficiency compared to the method of planning sequences performed by skilled researchers aided by algorithms. To

Fig. 5 Time diagram of different disassembly sequence planning methods



further improve the efficiency of power battery disassembly sequence planning, the following steps can be taken:

(1) Establish a power battery disassembly information graph base and introduce power battery similarity comparison in the algorithm. By finding the power battery disassembly information model with the highest matching degree with the current battery in the power battery disassembly information model base, the algorithm can update and store it to improve the efficiency of disassembly sequence planning.

(2) Further improve the algorithm to enhance its compatibility for the data on the input side. The algorithm should be able to handle changes in disassembly methods, such as parallel disassembly of power batteries, collaborative human–machine disassembly, selective disassembly, etc.

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Date Availability Not applicable.

Code Availability Not applicable.

Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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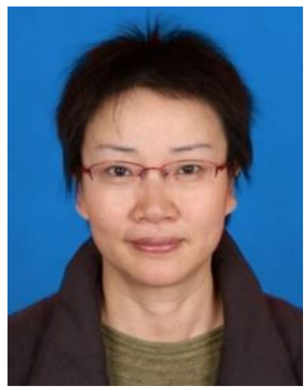

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